

Milestone 1: Production RAG System

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DATA 790

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Goal: Develop a production RAG system around the humble sweet potato. Users might include farmers, agricultural extension agents, agronomists, and home gardeners based on the current corpus of research articles, NC State Extension websites, and data tables from the U.S. Department of Agriculture surrounding genetic diversity, growing traits, nutritional value, and various pathogens and pests that affect sweet potato crops.

System

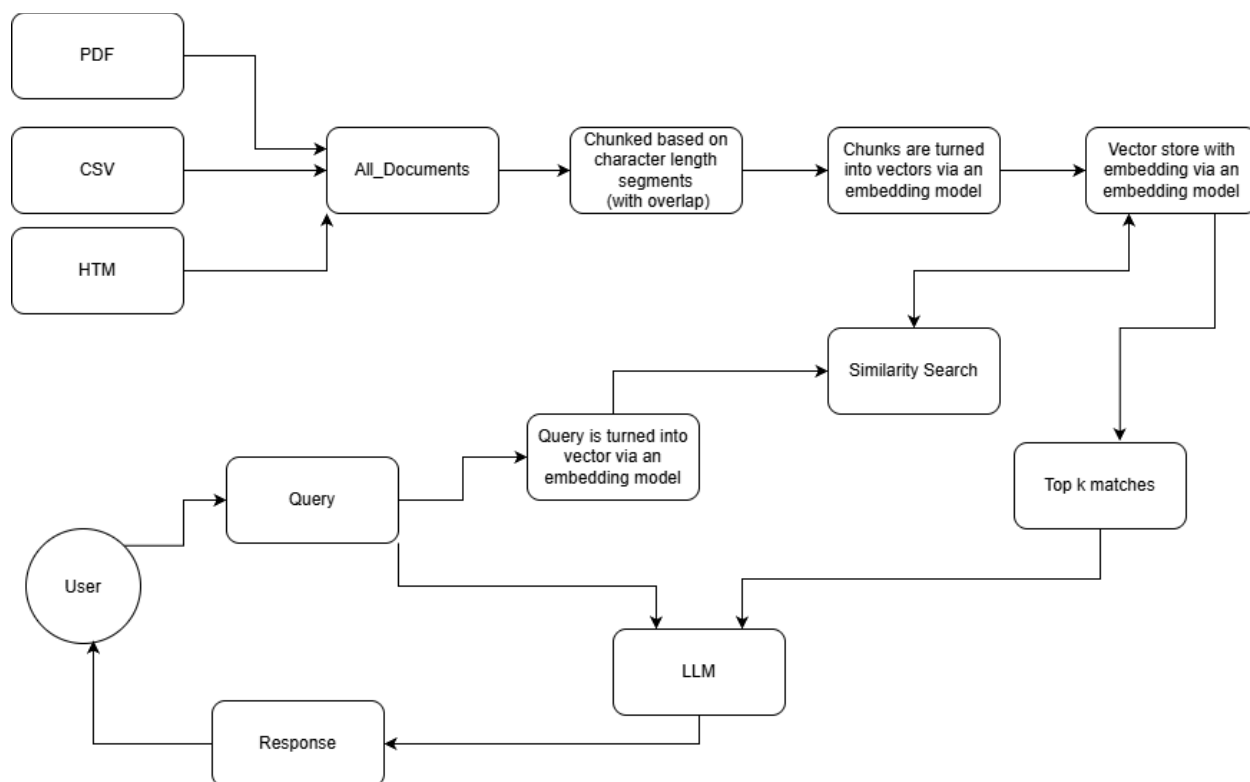
This system takes in pdf, csv, and htm files and throws an error if an unsupported file type is attempted. These documents go through some slight cleaning, especially the htm files, so that document type specific metadata can be assigned to sections if needed. Then the documents are loaded into one variable.

From there, the documents are chunked based on a character length with a given overlapping size. The strategies used for this milestone included:

- Small chunks: 500 character lengths with 50 character overlap
- Large chunks: 1000 character lengths with 50 character overlap
- Large chunks with greater overlap: 1000 character lengths with 200 character overlap

After chunking, these chunks were fed into an embedding model. The embedding model used was OpenAI's text-embedding-3-small. While this was the default model, it has a higher performance average and higher speed when compared to the other embedding model we have used in this course, text-embedding-ada-002.

Following embedding, the vectors are stored in a vector db. For the bare minimum RAG pipeline, Weaviate was used as a precursor to introduce hybrid search into the RAG. However, there were limited free collections available and the syntax to utilize one for multiple chunking strategies was detracting from completing the project. For the evaluation portion of this project, Chroma was used so that vector stores could be stored locally. This was possible due to the limited number of files utilized in the evaluation process. K was defaulted to 3 as set forth in previous RAG pipeline evaluations as the number of chunks that gives the highest performance without a continual increase in tokens for only marginal improvements. For the generation model, gpt-4.1-mini was utilized as the chat model. This model is significantly cheaper to use than the frontier model and has sufficient capability to generate a response given a query and relevant context. For the evaluation of query responses however, gpt-5.6-sol was utilized so that a higher tier model was acting as judge.



The Gold Set

For previous evaluations of RAG pipelines, a pdf only document based was developed and 10 questions with location identified via page index were written into a dictionary. Using the LLM chat model, additional questions were generated and evaluated for addition into a gold set. The final gold set for this project was 34 questions.

- 13 questions from the pdf files (original 10 via human and 3 generated)
- 11 from the csv files (all generated)
- 10 from the htm files (all generated)

With each question was stored the relevant metadata for the file type, such as source for all also page for pdf files, section index for htm files, and row for csv files.

It should be noted that the model had significant trouble generating semantic questions from the tabular csv files. While two csv files were in the corpus, the model was only able to generate questions from one of them. In addition, not included in this repository is a third csv that was attempted at introducing into the corpus. Not only were questions difficult to generate due to excessive use of acronyms in the column headers, but the results for questions that should have come from pdf were produced with chunks from this csv, claiming higher semantic similarity (a closer L2 distance) than the chunks from pdfs. This resulted in nonsense outputs. This particular file had to be removed and as the model was unable to generate questions

based on the third csv file, it was left in place but not formally assessed in this evaluation. Further discussion on file type follows.

Retrieval Evidence & Evaluation

Three chunking strategies were initially explored for this expanded document base and gold questions set with which to test retrieval. These were the same strategies used in prior experiments:

- Small chunks: 500 character lengths with 50 character overlap
- Large chunks: 1000 character lengths with 50 character overlap
- Large chunks with greater overlap: 1000 character lengths with 200 character overlap

The metrics recorded were hit rate, average recall, mean reciprocal rank, and precision. All strategies were run with the number of chunks held constant, $k = 3$. The results of this evaluation were:

strategy	n	hit_rate	recall	mrr	precision
Large Chunks (1000/50)	34	0.912	0.897	0.814	0.382
Large Chunks Larger Overlap (1000/200)	34	0.824	0.809	0.770	0.363
Small Chunks (500/50)	34	0.794	0.794	0.735	0.353

Large Chunks (1000/50) is the clear winner across all key retrieval metrics, achieving a 91.2% hit rate and a mean reciprocal rank (MRR) of 0.814. Dropping chunk size from 1000 to 500 dropped the hit rate by nearly 12 percentage points (0.912 to 0.794) and MRR dropped by 0.079. This is most likely due to the technical content found within the research articles. Plant pathology and soil management have highly specific domain semantics that require additional explanatory blocks to preserve meaning. Within text that has cause and effect or symptoms and treatments, the smaller chunking blocks are more likely to separate the important connections. This leads to weaker vector embeddings and missing key terms during retrieval.

Expanding the overlap with the same chunking size, (1000/50 vs. 1000/200) decreased the hit rate by 8.8 percentage points and degraded MRR from 0.814 to 0.770. Based on the understanding of the issue with smaller chunks, at first glance, more overlap seems like it should be better. However, with greater overlap, the vectors created near each other hold greater similarity. This creates redundancy, true, but then the distinct chunks needed for a particular query will be crowded out.

Utilizing large chunks (1000/50) leads to higher MMR which explains that when the correct chunk is found, it is ranked higher compared to other chunks selected from distance similarity. With the large chunk strategy 0.814 for the winning strategy indicates that when relevant context is found, the single best matching chunk is ranked first or second in the retrieval results the vast majority of the time.

The precision values are low but expected. If only $k = 3$ chunks were evaluated, the system often retrieved the correct location somewhere in the top 3, but the other retrieved chunks were frequently not gold locations. As most questions were only tied to one subset of one file, these low precision scores are to be expected. A greater enriched goldset with questions referring to multiple chunks would better assess the precision metric.

Overall, these scores were lower than a previous experiment that only evaluated retrieval from a pdf only document base. It was decided to dig deeper into the metrics to evaluate if there was a difference based on file type, especially with the highly structured aspect of csv files and highly unstructured aspects of htm and pdf files. The results are as follows:

		n	hit_rate	recall	mrr	\
filetype	strategy					
csv	Large Chunks (1000/50)	11	0.818	0.818	0.818	
	Large Chunks Larger Overlap (1000/200)	11	0.727	0.727	0.727	
	Small Chunks (500/50)	11	0.818	0.818	0.818	
htm	Large Chunks (1000/50)	10	0.900	0.900	0.800	
	Large Chunks Larger Overlap (1000/200)	10	0.800	0.800	0.683	
	Small Chunks (500/50)	10	0.700	0.700	0.650	
pdf	Large Chunks (1000/50)	13	1.000	0.962	0.821	
	Large Chunks Larger Overlap (1000/200)	13	0.923	0.885	0.872	
	Small Chunks (500/50)	13	0.846	0.846	0.731	

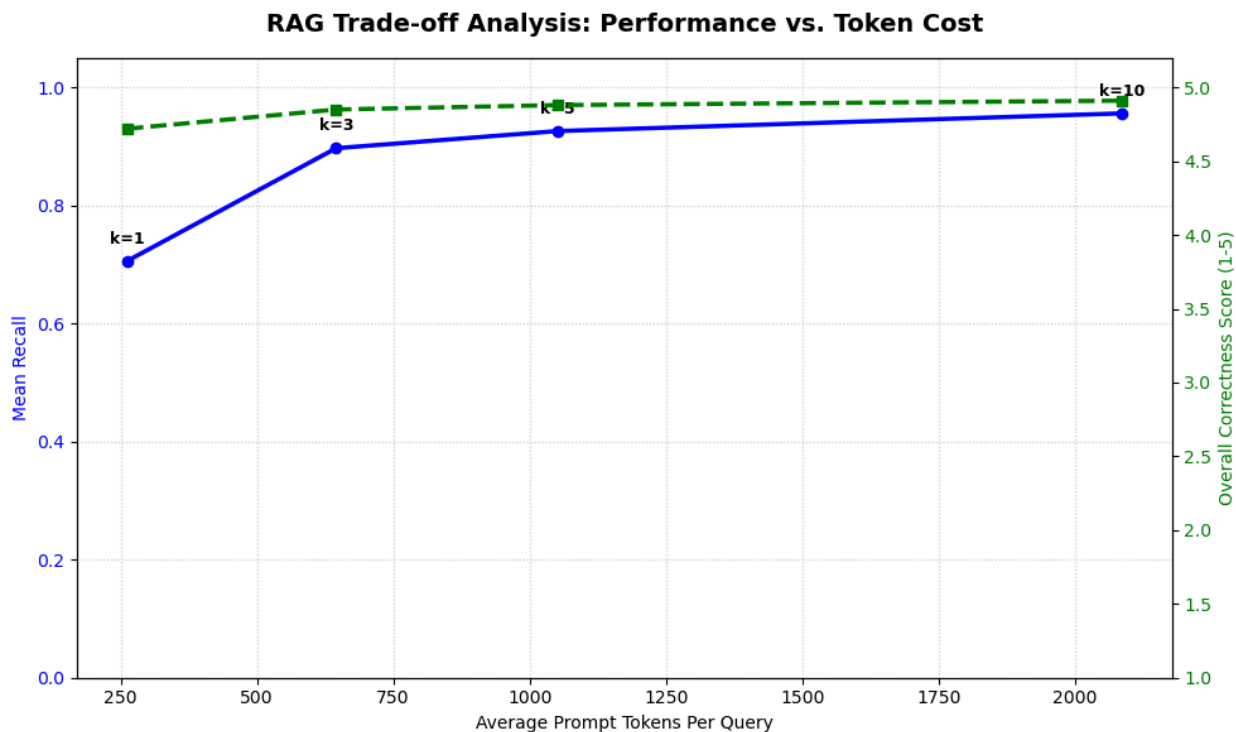
		precision
filetype	strategy	
csv	Large Chunks (1000/50)	0.273
	Large Chunks Larger Overlap (1000/200)	0.242
	Small Chunks (500/50)	0.273
htm	Large Chunks (1000/50)	0.400
	Large Chunks Larger Overlap (1000/200)	0.333
	Small Chunks (500/50)	0.367
pdf	Large Chunks (1000/50)	0.462
	Large Chunks Larger Overlap (1000/200)	0.487
	Small Chunks (500/50)	0.410

PDFs are still a top performer with 100% hit rate with the large chunks strategy. The narrative nature of these mostly academic papers allows for the chunks to preserve contextual meaning without splitting critical details. CSV files performed the same in both large and small chunk strategies. With fixed sized chunking, chunking can happen in the middle of rows, which disrupts relationships between columns. Increasing the tokens does not fix this problem and instead just increases the number of rows in a disconnected chunk. The low precision (0.273) reflects that retrieved csv chunks contain noise from surrounding unrelated rows. HTM files follow a similar trend to pdfs and large chunks (1000/50) still wins with a 0.900 hit rate and 0.800 MRR. These specific htm files were of a similar academic nature to the pdfs and included narrative prose. However, there were also tables, menus, inline tags, references to images without the image, and other noise to the narration. This can disrupt the semantic meaning.

While some attempt was made to clean the noise, there is a slight dip in retrieval metrics compared to cleaner pdfs. Chunking performance varied by source type. It does appear that heterogeneous document bases can affect overall retrieval metrics. Larger chunks performed particularly well for pdf and htm retrieval, while csv retrieval showed stronger performance with smaller chunks or greater overlap. There may not be one universally optimal chunking strategy across document types.

The above evaluation was done with $k = 3$ chunks as it was determined to be the turning point for quality and cost tradeoff in a prior, pdf only evaluation. K-tuning was repeated here with $k = 1, 3, 5$, and 10 . The function for k-tuning involves using an LLM-as-judge for additional metrics such as faithfulness, relevance, and overall correctness. More details on this strategy follow the results of k-tuning. The results of k-tuning are as follows:

k	Avg Prompt Tokens	Mean Recall	Faithfulness (1-5)	Relevance (1-5)	Overall Correctness	Total Cost (\$)
1	262.2	0.706	4.65	4.79	4.72	0.007574
3	645.2	0.897	4.82	4.88	4.85	0.013727
5	1050.7	0.926	4.76	5.00	4.88	0.015524
10	2086.1	0.956	4.85	4.97	4.91	0.021128



Going from $k = 1$ to $k = 3$ resulted in substantial improvement across the metrics, but did increase the token amounts and cost. There was roughly 84% increase in cost per query in this experiment, but retrieval recall jumps by about 28.4 percentage points. Moving from $k = 3$ to $k = 5$, the costs and token increases only give marginal increases in the metrics. That pattern continued moving from $k = 5$ to $k = 10$. Overall, increasing k improved retrieval recall, but the gains diminished as more documents were retrieved. Smaller k values substantially reduced

prompt-token usage and cost, while $k = 10$ provided the highest recall and generation scores in this evaluation. The remainder of the project was kept at $k = 3$ to be consistent with past experiments and this round of chunking strategy evaluation.

To evaluate the generation that resulted from the input of the query and selected context via dense vector search, the strategy of LLM-as-Judge was used. A chat level model was used to evaluate the query and selected context and return a generated response. A frontier model was then given the query, selected context, and the generated response to evaluate faithfulness and relevance as instructed via a judge specific prompt. Faithfulness was scaled from 1 = mostly unsupported or invented to 5 = fully supported) based on the claims in the generated response supported by the retrieved context. Relevance was scaled from 1 = off-topic to 5 = fully answers it based on if the generated answer directly addresses the question. These scales were applied to all of the 34 questions in the gold data set and then averaged.

The results of the LLM judge are as follows:

- Mean Faithfulness: 4.79
- Mean Relevance: 4.97

Faithfulness scoring 4.79 / 5.00 leads to the conclusion that there is a low hallucination rate. The chat level model is not creating context to fill in gaps and instead is supporting the generation of the response with the context retrieved from the similarity search. Relevance scoring 4.97 / 5.00, almost perfectly, indicates that responses were direct and completely addressed the specific question. There is a low likelihood of partial answers or irrelevant filler text. What was not evaluated and should be in future study is faithfulness and relevance scores between file types.

Retrieval Failures

Questions that got 0 hits for a particular strategy and file type were recorded. One question, "What has led to the reemergence of black rot in North Carolina sweet potatoes?", came from the original, human made gold set of questions. The expected information was located in rag_sp2.pdf page 6 and rag_sp4.pdf page 12. However, with the small chunks (500/50) strategy, all three retrieved results came from rag_sp2.pdf page 0, so neither of the expected source locations was retrieved. This indicates a retrieval failure rather than a generation failure, because the relevant passages were not included in the retrieved context. The chunk size and overlap could be adjusted to address this issue. Increasing chunk size and/or overlap may help preserve related sentences or paragraphs within the same retrievable chunk. Increasing k could also be evaluated, although modifying the chunking strategy may be more appropriate because the relevant passages were not retrieved at all.

Strategy: Small Chunks (500/50)
File Type: pdf

QUESTION:

What has lead to the reemergence of black rot in North Carolina sweet potatoes?

EXPECTED:

```
[{'source': 'rag_sp2.pdf', 'page': 6}, {'source': 'rag_sp4.pdf', 'page': 12}]
```

RETRIEVED:

```
{'source': 'rag_sp2.pdf', 'page': 0, 'row': None, 'section_idx': None}  
{'source': 'rag_sp2.pdf', 'page': 0, 'row': None, 'section_idx': None}  
{'source': 'rag_sp2.pdf', 'page': 0, 'row': None, 'section_idx': None}
```

For the question, “For the plant known as Daxua with the GRIN Database PI Accession PI 315349, what is the width of its leaf in centimeters?”, the expected record was sp_table2.csv, row 27. With the Large Chunks Larger Overlap (1000/200) strategy, the system instead retrieved rows 680, 189, and 187. This also indicates a retrieval failure, since the correct csv row was not included in the retrieved context. For this type of structured data, the csv ingestion and chunking approach could be modified rather than simply increasing character chunk size or overlap. Fixed character based chunking may separate related fields or group unrelated records together. Converting each csv row into a descriptive key-value representation before embedding could keep the plant name, accession number, and leaf-width measurement together, potentially improving retrieval of the correct record.

Strategy: Large Chunks Larger Overlap (1000/200)
File Type: csv

QUESTION:

For the plant known as Daxua with the GRIN Database PI Accession PI 315349, what is the width of its leaf in centimeters?

EXPECTED:

```
[{'source': 'sp_table2.csv', 'row': 27}]
```

RETRIEVED:

```
{'source': 'sp_table2.csv', 'page': None, 'row': 680, 'section_idx': None}  
{'source': 'sp_table2.csv', 'page': None, 'row': 189, 'section_idx': None}  
{'source': 'sp_table2.csv', 'page': None, 'row': 187, 'section_idx': None}
```

The current system uses Chroma which utilizes dense vector similarity retrieval with a default distance calculation of L2. A future version should integrate hybrid retrieval by combining the existing semantic retriever with a BM25 keyword-based retriever using LangChain or use a vector store such as Weaviate that includes reranking within the search algorithm. This could improve retrieval for queries containing specific terminology, names, accession numbers, or other exact terms that may not be captured as effectively through semantic similarity alone. Weaviate was the best at MRR on a pdf only document base in prior experiments and should be investigated for this growing, multi-file type base.

Cost

A cost tracker and cost calculator were introduced to the RAG pipeline. These track prompt tokens, completion tokens, latency, total cost via a pricing database and partial abstraction to the pricing database so that changes on future costs and model availability can be updated. The cost tracker was first applied to the chunking strategy outcomes

--- Cost & Performance Efficiency Comparison ---

Strategy	Avg Prompt Tokens	Avg Completion Tokens	Avg Total Tokens	Avg Latency (s)	Total Cost (\$)
Small Chunks (500/50)	438.68	68.06	506.74	1.518	0.009668
Large Chunks (1000/50)	645.24	91.76	737.00	1.688	0.013767
Large Chunks High Overlap (1000/200)	662.97	85.53	748.50	1.741	0.013669

Unsurprisingly, when the larger chunked vectors are included as context more tokens are required and latency increases. However, these costs should be balanced against increases in metrics. Further granularity regarding costs based on file type should be explored.

Because abstraction was added to this experiment, the ability to select the chat model is now available. Chat model selection varies based on the parameters set forth by those hosting this RAG pipeline, but costs are often a key constraint.

Cost Projections (Monthly):

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Small App (1,000 requests/day):

gpt-4o-mini	: \$	5.85/month
claude-4.5-haiku	: \$	45.00/month
gpt-4o	: \$	97.50/month

Medium App (10,000 requests/day):

gpt-4o-mini	: \$	58.50/month
claude-4.5-haiku	: \$	450.00/month
gpt-4o	: \$	975.00/month

Large App (100,000 requests/day):

gpt-4o-mini	: \$	585.00/month
claude-4.5-haiku	: \$	4,500.00/month
gpt-4o	: \$	9,750.00/month

The chat model gpt-4o-mini was chosen due to the quality of its responses to direct questions, often able to be answered with one chunk of text as demonstrated with the gold question set. However, if a higher tier model would be desired, the projected monthly costs are shown to the left. These are the costs for the same 34 question gold set averaged but applied at scale based on the number of users utilizing the app supported by this RAG pipeline and the model chosen by the hosts. It would be projected that introducing hybrid search and with more advanced queries that require references to multiple chunks across document sources would increase the costs that are shown here.

But the trade offs in capability of the RAG pipeline should be evaluated in a future cost-benefit analysis study.

Costs for a RAG pipeline can be optimized with selected strategies such as prompt trimming, difficulty-based routing, and caching. At the current level of performance, the proposed optimization strategy would be caching. As the collection of documents is highly specialized to sweet potatoes, users may repeatedly ask questions about the same sweet potato diseases, management practices, or varieties. Semantically similar questions could reuse an existing answer without making another LLM call, reducing both token usage and latency. This approach could also preserve the existing retrieval and generation pipeline rather than reducing the

amount of contextual information provided to the model. A semantic caching strategy was considered but was not implemented or included in the final cost measurements.

Limitations

Future retrieval improvements could be made through how different file types are ingested or chunked. One such option is to not run fixed character text splitters on csv files and instead iterate through rows and convert each row into a descriptive text sentence or key-value format. Further cleaning practices for htm files should also be explored. Further study on the effect of different file types on various retrieval and generation metrics should also be conducted, including their impact on cost. While retrieval scores were high across file types and chunking strategies, the number of documents and questions linked to only one gold chunk are limitations. Expansion of the corpus, adding hybrid search to support more complex and exact-match queries, and evaluating retrieval performance with multiple relevant chunks per question would provide a more comprehensive assessment of system performance.

Future work should also address security and production safeguards. Prompt injection testing should be incorporated to evaluate whether malicious or misleading instructions within user queries or retrieved documents can alter system behavior. Access controls should be investigated to ensure that users can retrieve only information they are authorized to access, particularly if the system is expanded to include non-public documents. Additional guardrails and input validation could also be implemented to detect unsafe or inappropriate inputs and constrain generated responses to the retrieved evidence. These measures would help evaluate the system beyond retrieval accuracy and move it toward a more robust production-ready RAG application.

Sources for RAG

HTM Files:

Quesada-Ocampo, Lina, and Madison Stahr. "Geotrichum Sour Rot of Sweetpotato: NC State Extension Publications." *Geotrichum Sour Rot of Sweetpotato | NC State Extension Publications*, 24 Sept. 2018, content.ces.ncsu.edu/geotrichum-sour-rot-of-sweetpotato.

Quesada-Ocampo, Lina, and Madison Stahr. "Southern Blight of SWEETPOTATO: NC State Extension Publications." *Southern Blight of Sweetpotato | NC State Extension Publications*, 26 Sept. 2018, content.ces.ncsu.edu/southern-blight-of-sweetpotato.

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Quesada-Ocampo, Lina. "Sweetpotato Root Knot Nematode: NC State Extension Publications." *Sweetpotato Root Knot Nematode | NC State Extension Publications*, 24 May 2018, content.ces.ncsu.edu/sweetpotato-root-knot-nematode.

Quesada-Ocampo, Lina. "Sweetpotato Scurf: NC State Extension Publications." *Sweetpotato Scurf | NC State Extension Publications*, 16 Dec. 2018, content.ces.ncsu.edu/sweetpotato-scurf.

CSV Files

Data from: Simultaneous quantification of β -carotene, anthocyanins, and total phenolics in sweetpotato with visible-near-infrared spectroscopy

Published by Agricultural Research Service | Department of Agriculture | Catalog Last Checked: September 10, 2026 at 07:50 PM | Dataset Last Updated: January 09, 2026

– Sweetpotato Phytonutrients Contents.csv

https://catalog.data.gov/dataset/data-from-simultaneous-quantification-of-carotene-anthocyanins-and-total-phenolics-in-sweetpotato-with-visible-near-infrared-spectroscopy?from_hint=eyJxIjoic3dlZXRwb3RhdG8ifQ%3D%3D

Data from: Phenotypic Variation in Leaf Morphology of the USDA-ARS Sweetpotato Germplasm Collection

Published by Agricultural Research Service | Department of Agriculture | Catalog Last Checked: September 10, 2026 at 07:50 PM | Dataset Last Updated: February 13, 2024

– Sweetpotato-Leaf-Descriptor-Summary Table.csv

https://catalog.data.gov/dataset/data-from-phenotypic-variation-in-leaf-morphology-of-the-usda-ars-sweetpotato-germplasm-collection?from_hint=eyJxIjoic3dlZXRwb3RhdG8ifQ%3D%3D

PDFs

Burri, Betty J. "Evaluating sweet potato as an intervention food to prevent vitamin A deficiency." *Comprehensive Reviews in Food Science and Food Safety*, vol. 10, no. 2, 18 Feb. 2011, pp. 118–130, <https://doi.org/10.1111/j.1541-4337.2010.00146.x>.

George, Justin, et al. "Sustainable sweetpotato production in the United States: Current status, Challenges, and opportunities." *Agronomy Journal*, vol. 116, no. 2, Feb. 2024, pp. 630–660, <https://doi.org/10.1002/agj2.21539>.

George, Nicholas A., et al. "The carbohydrate yield of Sweetpotato (*Ipomoea batatas*) grown from slips and root pieces in North Carolina." *HortScience*, vol. 50, no. 11, Nov. 2015, pp. 1610–1617, <https://doi.org/10.21273/hortsci.50.11.1610>.

Jonson, Taylor. *Commodity Highlight: Sweet Potatoes*, United States Department of Agriculture, 1 May 2015, ers.usda.gov/media/17105/commodity-highlight-sweet-potatoes.pdf?v=35139.

Scruggs, A. C., et al. "Genetic Diversity, fungicide sensitivity, and host resistance to *ceratocystis fimbriata* infecting sweetpotato in North Carolina." *Plant Disease*, vol. 101, no. 6, June 2017, pp. 994–1001, <https://doi.org/10.1094/pdis-11-16-1583-re>.

Wadl, Phillip A., Bode A. Olukolu, et al. "Genetic diversity and population structure of the USDA Sweetpotato (*ipomoea batatas*) germplasm collections using GBSpoly." *Frontiers in Plant Science*, vol. 9, 21 Aug. 2018, <https://doi.org/10.3389/fpls.2018.01166>.

Wadl, Phillip A., William B. Rutter, et al. "USDA 'cascade': A multiple pest-resistant, rose-skinned, creamy yellow-fleshed Sweetpotato." *HortScience*, vol. 60, no. 10, Oct. 2025, pp. 1772–1774, <https://doi.org/10.21273/hortsci18853-25>.

AI Conversation Links

<https://claude.ai/share/49702a5f-3dfc-48fa-9d53-5095fef249f2>
<https://claude.ai/share/bc2a2d08-f441-4d9f-8dcc-879f6791267e>
<https://claude.ai/share/f66d3aec-5df9-464a-b4ed-90614e4a60ac>
<https://claude.ai/share/9ed69b92-fa73-4839-b85c-85ca545fa343>
<https://claude.ai/share/2be6f0c2-e0b8-4e15-b05a-e190fddf1c99>
<https://claude.ai/share/6e3b842d-260d-483a-b8be-5c6697669a3c>
<https://chatgpt.com/share/6abcaa73-c110-83ea-91e2-102ad398aede>